



AI, autism, and the architecture of voice: from engineered exclusion to designed dignity

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Abstract

This paper conceptualizes *engineered exclusion*—the predictable sidelining of disabled users resulting from choices about data provenance, model objectives, and evaluation practices within AI systems. We examine engineered exclusion through the lived experiences of minimally and nonspeaking autistic people whose communicative profiles challenge the speech-centered defaults embedded in contemporary AI pipelines. Here, “voice” refers not only to speech but to the broader architecture through which embodied, multimodal communication—spanning AAC text, gesture, movement, and partial vocalizations—becomes legible within AI systems. “Nonspeaking” is therefore not the absence of language but a heterogeneous spectrum in which communication is often state dependent, varying with fatigue, anxiety, sensory load, and motor planning demands—forms of variation that design abstractions routinely erase. Tracing exclusionary mechanisms across speech recognition, text-to-speech, plain-language systems, and interface design, we introduce measurable *designed-dignity* metrics for technical evaluation (Table 1) and a governance framework mapping accountability across the AI lifecycle (Table 2). We argue that accessibility must be treated as a core dimension of AI ethics—on par with fairness, privacy, and safety. Re-engineering AI for designed dignity requires systems that recognize embodied, multimodal, and fluctuating forms of communication, expanding what counts as valid signal and responsible innovation.

Keywords AAC · Accessibility · AI ethics · Autism · Communication · Designed dignity · Engineered exclusion

Artificial intelligence (AI) increasingly mediates how people learn, work, and communicate, shaping access to information and participation in social life through tools such as predictive text, voice assistants, automated translation, and plain-language summarizers. Yet for many minimally and nonspeaking autistics, individuals who cannot rely on speech alone to be heard and understood (Communication-FIRST, n.d. 2026), the same technologies reproduce long-standing patterns of exclusion, especially for communication that is variable, multimodal, or state dependent. This paper introduces engineered exclusion, the predictable sidelining of disabled users that arises from choices about data provenance, model objectives, and evaluation practices in contemporary AI pipelines. Our focus is squarely on AI systems and development processes; we do not engage in broader social debates about terminology or identity except where

they directly affect design and evaluation. While the analysis centers on speech- and language-based systems—such as speech recognition, text-to-speech, and AAC interfaces—the conceptual frame of *engineered exclusion* also extends to generative text systems that mediate communication access for users with motor or linguistic challenges. These systems, too, encode assumptions about fluency, predictability, and “legible” expression that can marginalize disabled communicators (Bender et al. 2021; Freyer 2024). Documented disparities in speech technologies and the predominance of “speaking,” native-English user profiles in datasets and design assumptions mean minimally or nonspeaking autistics are often misclassified or omitted, patterns that mirror historical exclusions regarding whose communication is deemed valid (Koencke et al. 2020; Srinivasan 2025b).

This paper makes three contributions: (1) it frames accessibility as a first-order AI ethics concern rather than a compliance add-on; (2) it proposes operationalizable designed-dignity metrics (Table 1) that translate critical disability scholarship into engineering-relevant checkpoints; and (3) it centers minimally and nonspeaking autistic communicators,

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Table 1 Operational mechanisms of engineered exclusion across modalities and corresponding designed-dignity metrics

Domain	Engineered mechanism	How exclusion operates	Designed-dignity metric
Automatic speech recognition (ASR)	Distributional underrepresentation	Default training corpora underrepresent atypical, dysarthric, accented, or state-variable speech. Fine-tuned models exist but are not baseline deployment behavior	Inclusion-weighted dataset audits; WER parity reporting across speech variability profiles; baseline deployment transparency
	Enrollment and adaptation dependency	Performance improvements require personalized fine-tuning, sufficient training samples, or stable articulation—placing adaptation burden on user	Default-on adaptive modeling; low-sample personalization benchmarks; fatigue-sensitive evaluation protocols
	Ecosystem dependence	Accessibility improvements may require specific devices, updates, or supported languages	Cross-platform accessibility parity; multilingual atypical speech coverage reporting
	Conditional Identity access	Identity-aligned voices embedded within hardware ecosystems, premium tiers, or technically complex workflows rather than universal default	Identity-fit availability index; hardware-independent voice portability; expressive-fidelity baseline standards
	Enrollment burden	Voice cloning requires sustained recording sessions, articulation stability, motor endurance, and device navigation capacity	Enrollment effort audits; motor-access adaptive enrollment modes; articulation-variability tolerance testing
	Grapheme-to-phoneme (G2P) normalization bias	Acronyms, scientific terms, multilingual names, and non-dominant pronunciations default to majority-frequency inference, requiring phonetic workaround labor	Domain-specific pronunciation fidelity testing; acronym handling benchmarks; user-controlled pronunciation override tools
Cross-modal Infrastructure	Correction labor transfer	Users must pre-correct spelling, spacing, pacing, or phonetics to force accurate output	Cognitive-labor impact metrics; automated contextual pronunciation adaptation; expressive-latency tracking
	Priority-driven remediation	Highly visible failures corrected rapidly; subtle representational inequities persist longer	Equity-prioritization review criteria; time-to-remediation disparity tracking
	Majority-frequency optimization	Systems treat statistical frequency as correctness, defining normative expression as baseline	Variability-weighted evaluation metrics; minority-expression fidelity thresholds

The table summarizes structural mechanisms through which AI systems embed exclusion and proposes corresponding measurable indicators for designed dignity. These metrics are illustrative and intended to guide operationalization across modalities. These metrics are illustrative and intended to guide operationalization; their validity and reliability require empirical evaluation across diverse user populations and deployment settings

a population largely absent from mainstream AI fairness discourse. We also offer a concise governance map for policy-makers to identify where accountability should attach across the AI lifecycle (Table 2).

Crucially, “nonspeaking” or “minimally speaking” is not a single condition or an absence of language. It names a wide spectrum of communication and users of augmentative and alternative communication (AAC): some people type on keyboards (with varying fluency and with the use of a single index finger to using all 10 fingers), others point to letterboards, symbol-based systems, or write on paper. Many also communicate via gesture, rhythmic movement, partial vocalizations, or echolalia, repetition that can carry pragmatic meaning such as affirmation, regulation, or refusal (Prizant et al. 2006). AAC refers to the systems and technologies that support or replace natural speech, from no-tech letterboards and laminated picture symbols to high-tech speech-generating devices and apps, used to express and comprehend language across contexts (Beukelman and Light 2020).

Many autistics may be multimodal, combining varying degrees of speech and AAC usage depending on context. Within the same person, communicative profiles can vary across time and setting with fatigue, anxiety, sensory load, predictability, and motor planning demands (Tager-Flusberg and Kasari 2013). AAC use cuts across the DSM-5 levels in autism, though a substantial share of AAC users also meet the higher support Level 3 criteria, reflecting needs across communication, self-regulation, and daily living (American Psychiatric Association 2022; Beukelman and Light 2020). This overlap does not make the categories coextensive; moreover, support needs are state -dependent and can intensify with fatigue, overload, or unfamiliar environments while easing with predictable routines and supports.

A related consideration is motor planning. Many minimally or nonspeaking autistics contend with differences in planning and sequencing movement, often described clinically as apraxia or dyspraxia, alongside atypical sensory integration (Dziuk et al. 2007). These differences do not preclude intentional communication; they change how intention is expressed. Producing an utterance, text, pointed, gestured, or vocalized, may require conscious sequencing and control of movement, so pace and rhythm diverge from spontaneous speech even when the message is clear (Torres et al. 2013).

This heterogeneity, spanning both expressive rhythm and mutual understanding, matters because design abstractions tend to erase it. For AI design, the implication is straightforward: systems must center expressive access and tolerate variability, rather than presume a single, stable communicative profile. This misalignment resembles what Milton (2012) describes as the *double empathy problem*—a breakdown in mutual understanding between differently minded

Table 2 Governance sites across the AI lifecycle: from engineered exclusion to designed dignity

AI lifecycle stage	Dominant governance failure	Dignity-oriented governance question
Data provenance	Disabled communication underrepresented, medicalized, or distributionally marginalized in data collection	Who decides what counts as valid signal at data collection, and whose communication is systematically excluded or reframed?
Model training and optimization	Variability optimized away as statistical noise; majority-frequency patterns treated as normative baseline	How is communicative diversity weighted during optimization, and what variability is preserved rather than suppressed?
Interface and configuration design	Control and configuration centralized in caregivers, clinicians, or platform defaults; enrollment burden placed on users	Who holds agency over configuration, correction, and identity representation, and how is enrollment effort distributed?
Evaluation and benchmarking	Static, one-shot benchmarks that ignore variability, adaptation burden, and state-dependent performance	What forms of change, fluctuation, and adaptive use are recognized as success, and how is effort accounted for in evaluation?
Deployment and oversight	Limited public accessibility transparency; inclusion treated as feature rather than enforceable baseline	How is communicative inclusion made visible, measurable, and enforceable across deployment contexts?

The table identifies governance sites across the AI lifecycle where design and policy decisions systematically enable or constrain communicative dignity. It establishes the normative governance framework within which engineered exclusion can be identified and addressed, highlighting concrete points where oversight, accountability, and design interventions can shift AI systems toward dignity-oriented inclusion

communicators. In the same way that autistic and non-autistic interlocutors can misinterpret one another's signals, the "mind" of AI, trained on normative language and rhythm, can misread the embodied, nonlinear, or intermittent expressions of minimally speaking users. What results is a form of *algorithmic double empathy failure*: the model assumes the user is incoherent, while the user experiences the model as unresponsive. Both "sides" operate from incompatible assumptions about what communication should look like. Reframing engineered exclusion in this way highlights that accessibility is not simply a matter of adding more data, but of fostering systems capable of *reciprocal understanding*—systems that learn to adapt to diverse communicative logics rather than forcing users into legibility.

Engineered exclusion should therefore be understood as a dynamic process rather than a fixed condition. AI systems evolve rapidly, and visible product-level failures may be partially remediated over time. Developments in personalized voice synthesis, expanded atypical speech datasets, and improved generative interfaces demonstrate that specific breakdowns can be addressed when prioritized. Yet remediation at the product layer does not automatically alter the training distributions, evaluation benchmarks, economic incentives, or governance structures that define normative communication. Distinguishing product-level remediation from pipeline-level structural redesign clarifies how exclusion can appear reduced in particular tools while persisting infrastructurally, leaving improvements contingent rather than default.

By situating accessibility within the same technical and policy infrastructures that govern fairness, privacy, and safety, this paper aligns with and extends work in AI ethics (Benjamin 2019; Costanza-Chock 2020). We argue that creating AI systems usable by minimal and nonspeaking people is not only a moral imperative but also a methodological one, expanding what counts as valid signal, legitimate data, and responsible innovation. In doing so, it invites a re-engineering of AI pipelines toward designed dignity rather than engineered exclusion. *Designed dignity* refers to the intentional embedding of respect, agency, and expressive equity into the architecture of technology itself. Where accessibility retrofits systems after exclusion has occurred, designed dignity anticipates human variation from the outset, ensuring that participation is neither conditional nor extractive. It shifts the goal from simply enabling function to affirming personhood, treating disabled users not as exceptions to accommodate but as co-designers whose modes of communication expand what technology can recognize as intelligence or intent (Costanza-Chock 2020; Hamraie 2017).

Engineered exclusion differs from adjacent concepts in AI ethics such as algorithmic bias or algorithmic oppression (Benjamin 2019) in that it does not primarily describe disparate error rates or representational harm across demographic

groups. Rather, it names a patterned infrastructural misalignment between system assumptions and communicative forms that fall outside dominant data distributions. While bias frameworks often focus on inequitable outputs across predefined categories, engineered exclusion highlights how entire communicative modalities—multimodal, state-dependent, AAC-mediated—may be structurally unrecognized at the level of training data, model objectives, and evaluation benchmarks. In this sense, it extends but is not reducible to fairness discourse, emphasizing pipeline architecture rather than outcome disparities alone (Benjamin 2019; Whittaker et al. 2021).

The sections that follow trace how this erasure propagates through specific technologies; input–output priorities, voice identity, recognition, analytics, fairness framings, plain-language generation, and interface agency. The point is simple: any pipeline that assumes a single, stable communicative profile will systematically underserve people whose expression is embodied, multimodal, and state dependent. This technological bias mirrors a social one: just as workplaces and institutions often presume neurotypical modes of communication as the default, favoring fluent, rapid, and verbally consistent interaction, AI systems replicate that presumption in code. Both collapse communicative diversity into a narrow ideal of legibility, overlooking how expression naturally fluctuates with sensory, emotional, and contextual states (Heasman and Gillespie 2018; Milton 2012).

This analysis is conceptual rather than empirical. It draws on AI system documentation, benchmarking practices, disability scholarship, and technical literature on ASR (automatic speech recognition), TTS (text to speech), and AAC systems to identify recurring patterns of exclusion that cut across these domains. The proposed designed-dignity metrics are therefore best understood as operational hypotheses—ways of translating normative commitments into constructs that can be tested, refined, and evaluated within real-world systems.

The paper proceeds in two stages. First, it identifies domain-specific mechanisms through which exclusion becomes infrastructural across speech recognition and assistive voice systems, translating these into operational designed-dignity metrics (Table 1). Second, it shifts to governance, mapping how accountability must attach across the AI lifecycle—from data provenance to deployment oversight—if communicative dignity is to be structurally embedded rather than retrofitted (Table 2). These sections move from diagnosis to institutional design.

1 Engineered exclusion in practice

Artificial intelligence does not merely mirror human experience; it also helps define what counts as communication and, by extension, whose experiences are legible within it. Across

speech, language, and communication technologies, choices about data provenance, modeling targets, and evaluation criteria establish implicit assumptions about the “normal” user and what counts as intelligible behavior. For minimally and nonspeaking autistics, exclusion is therefore not accidental but structurally embedded in the AI pipeline—from what data are collected to how system success is evaluated.

This section examines the mechanisms through which these design assumptions translate into engineered exclusion in practice. Table 1 operationalizes these mechanisms by identifying domain-specific sites of exclusion alongside corresponding designed-dignity metrics. Table 2 situates these mechanisms within the AI lifecycle, highlighting where communicative dignity is routinely lost and where accountability must attach. By specifying these domains as discrete categories, the framework converts a broad critique into operational levers for design and governance (Costanza-Chock 2020; Hamraie 2017).

To make this more concrete, consider how one of these metrics might be instantiated in practice. For example, one way to formalize this is through an Inclusion-weighted word error rate parity (IW-WERP) index, which evaluates whether ASR systems perform equitably across speech profiles:

$$IW - WERP = (WER_{\text{atypical}}) / (WER_{\text{normative}}),$$

where WER_{atypical} represents the word error rate for atypical or AAC-mediated speech, and $WER_{\text{normative}}$ represents the word error rate for normative speech on the same system. Rather than optimizing for aggregate accuracy alone, this formulation reframes performance in terms of parity—centering whether atypical users are disproportionately burdened by system error. For instance, a threshold such as $IW - WERP \leq 1.25$ could be used to indicate that atypical users are not disproportionately burdened by system errors. Systems exceeding this range would signal a breakdown in inclusion at the level of core functionality. This example is illustrative. Similar parity-based or fidelity-based formulations can be extended to other domains (e.g., TTS expressive fidelity or semantic preservation in text simplification). The broader aim is not to fix a single metric, but to show how dignity can be translated into measurable system constraints without collapsing communicative diversity into a single standard.

1.1 Input–output asymmetry

AI’s design priorities reveal whose voices are treated as central to communication systems. Nearly every major platform—from smartphones to virtual assistants—has been optimized for receiving speech (speech-to-text) rather than producing it (text-to-speech or symbol-to-speech) (Liu et al. 2022). Market indicators reflect this imbalance. The global

speech and voice recognition market was valued at USD 20.25 billion in 2023, with speech recognition accounting for 64.6% of total share and projected to reach USD 23.11 billion by 2030 (Grand View Research 2024). Comparable, disaggregated data for text-to-speech investment or feature benchmarking remain scarce, suggesting that listening technologies attract disproportionate investment while expressive technologies remain siloed as assistive rather than integral to communication design.

This asymmetry presumes a user who is already speaking, fluent, and machine-legible—a presumption that fails many minimally or nonspeaking autistics who rely on technology to produce communication outwardly. When expressive output is treated as an afterthought, participation itself becomes conditional. The pattern echoes a broader social dynamic in which assistive needs are prioritized only when they align with dominant user demands (Ellcessor 2016). During the early COVID-19 pandemic, for example, remote-work technologies became “magically possible overnight” when the able-bodied world required them, despite disabled people requesting such flexibility for decades (Srinivasan 2023).

For minimally and nonspeaking autistics, the consequences are tangible: fewer robust pathways for producing speech and slower progress on tools that center expressive access (Alper 2017; Wickenden 2011). The imbalance between input and output is therefore not only technical but epistemic, shaping whose communication becomes legible within AI systems. The mechanisms that follow show how priorities embedded in benchmarking culture and funding flows translate this asymmetry into engineered exclusion (see Table 1).

1.2 AAC voices and identity fit

For minimally and nonspeaking users who rely on text-to-speech (TTS) or other outward communication to participate in education, employment, and civic life, voice identity is central to communicative dignity: a mismatched or distorted voice can undermine social standing, while an identity-aligned one supports autonomy, confidence, and self-presentation (Wickenden 2011). Achieving that identity fit, however, is rarely straightforward. Beyond technical barriers, AAC users face additional financial, temporal, and cognitive costs that compound inequity—turning what should be a right to self-representation into a labor- and resource-intensive task (Alper 2017; Quinn et al. 2025; Wickenden 2011). These costs manifest in two primary ways: financial barriers that gate expressive access, and time and effort burdens that make communication labor intensive.

Recent developments complicate a simple paywall narrative. Apple’s Personal Voice now enables users to generate a synthetic voice from less than 15 min of on-device recording, integrated with Live Speech and compatible

AAC applications without additional subscription fees (Apple Inc 2024). ElevenLabs' Impact Program provides free voice-cloning licenses for individuals with permanent voice loss, blindness, or low vision (ElevenLabs 2025). Open-source text-to-speech models such as XTTS-v2 and FishAudio S1-mini allow cross-lingual voice cloning from short reference samples, expanding technical feasibility beyond proprietary systems (Coqui AI 2024; FishAudio 2025). These developments reduce direct licensing costs and demonstrate meaningful progress in identity-aligned voice synthesis.

However, reduced licensing fees do not eliminate structural access conditions. Voice cloning presupposes access to compatible hardware ecosystems, stable recording conditions, sufficient motor control for enrollment, and articulation patterns that can be reliably modeled. For example, Apple's Personal Voice requires users to record approximately 150 prompted phrases over a 15–20 min session while the device processes the model on charge. While brief for many users, such sustained vocal and motor effort may be fatiguing or infeasible for individuals with motor coordination differences, fluctuating speech clarity, respiratory challenges, or state-dependent articulation variability. Moreover, successful cloning presumes that the user's articulation is sufficiently stable to generate a usable model; highly variable or low-volume speech may not yield reliable synthesis without repeated sessions.

Enrollment, therefore, is not a trivial step; it assumes a level of stamina, consistency, and device navigation capacity that is unevenly distributed across disability populations. On-device systems remain embedded in costly hardware infrastructures, while open-source implementations require technical literacy, compute resources, or intermediary support. Identity fit therefore remains conditional rather than assumed as a baseline design.

1.2.1 Financial costs

Achieving a natural, age-appropriate, and culturally relevant AAC voice often entails additional cost. Voice options that convey individuality are often embedded within specific hardware ecosystems, premium service tiers, or technically complex workflows that create layered barriers to expressive access. Free or default voices tend to sound robotic or infantilizing, or rely on Western female accents familiar from Siri, Alexa, or Google Maps. For many low-income or non-native English users, authentic voice representation becomes economically inaccessible, forcing a choice between limited defaults and silence. Even "premium" voices often lack cultural nuance, mispronounce non-Western names and intonations, and thereby erase linguistic identity.

1.2.2 Time and effort costs

Identity fit is not only about accent or tone; it is also about fidelity of expression. Many TTS systems fail to handle even basic English usage, such as long vowels, acronyms, or scientific abbreviations. For instance, the long vowel "A" must often be typed phonetically as "yay" to sound correct, or it defaults to a short vowel sound. The ubiquitous ABA therapy that many autistic children are subjected to in childhood must be typed as "yay b yay" to pronounce correctly.

Acronym handling further exposes how narrow these systems' assumptions are. Common acronyms are often misread as words, which works for patterns like "NASA" and "FIFA" but "EEG" results in choked or nonsensical speech; users must add extra spaces between the letters to force proper enunciation. Since "EEG" contains long vowels, it must be typed phonetically as "ee ee g" to get legible audio output. If the acronym contains vowels, the system assumes it is a pronounceable word ("NASA," "FIFA"), whereas consonant-only acronyms ("VR") are treated like true acronyms and each letter is said separately. What seems like an arbitrary rule-based inference can be very confusing and frustrating for the user. Even then, this rule-based inference breaks down for mixed or domain-specific patterns. For example, AR (augmented reality) is rendered as "Arkansas," PT becomes "point," and the chemical symbol "Ca" for calcium becomes "California." Even standard phrases like "check-ins" default to "check inches." These are not edge cases; they are ordinary English expressions mispronounced because the system's grapheme-to-phoneme (G2P) and text normalization modules are tuned to a narrow subset of English usage (Deviyani 2022; Kuligowska et al. 2018; Reichel and Pfitzinger 2006). These front-end normalization modules, trained primarily on web-scale English text, default to the most frequent word associations, "correcting" rare or domain-specific terms into more probable but wrong pronunciations.

For AAC users, the cost of these design assumptions is steep. To be understood, they must pre-correct the system—adding apostrophes (e.g., base'd, liv'd), phonetic spellings, artificial pauses, or spelling out acronyms—to force accurate output. In effect, users must learn a complex meta-language simply to communicate, discovering through trial and error which spellings or formatting will produce intelligible speech. This workaround is not trivial: it adds layers of cognitive and motor labor to an already effortful process. Constructing a message letter by letter is slow enough; retyping words phonetically or experimenting with spellings multiplies that burden. Communication becomes a form of system debugging rather than self-expression. For minimally and nonspeaking users—whose every utterance already requires coordination of attention, motor planning, and intent—this additional layer

of correction can transform accessibility into exhaustion (Koencke et al. 2020).

These failures are not simply algorithmic accidents; they are predictable outcomes of how AI speech systems are built. Modern text-to-speech pipelines are trained on narrow, high-quality datasets dominated by fluent, native-English speakers—typically drawn from non-clinical, professionally recorded populations. Because such corpora exclude disordered or atypical speech by design, the resulting systems equate “clarity” with standardized pronunciation and prosody. “Naturalness” is then evaluated through human-rating metrics that reward fluency and familiarity, effectively defining “good” speech as what sounds most typical to the majority of listeners. This training ecology makes linguistic and articulatory diversity statistically undesirable, rendering many AAC-style utterances—typed, synthesized, or rhythmically patterned—less intelligible precisely because they deviate from the data distribution (Kuligowska et al. 2018; Prinos et al. 2024; Taylor and Richmond 2019).

These design choices reveal how TTS models implicitly define linguistic normalcy. By treating statistical frequency as correctness, AI voices mirror the linguistic dominance of high-resource English and the speakers who produce it. The result may sound fluent to outside listeners while distorting the intended meaning of those who depend on the system most. Voice identity, therefore, is not just an aesthetic preference—it marks the boundary between participation and exclusion. These financial and labor costs compound one another. As with ASR systems, recent technical advances show that identity-aligned synthesis is possible when prioritized. The structural question is whether such accommodation is treated as a universal baseline or remains contingent on enrollment, resources, and platform access. When customization or accuracy depends on ecosystem ownership, advanced configuration, or subscription tiers, inclusivity becomes conditional rather than guaranteed.

Paying for a voice that can correctly pronounce one’s own name or field-specific terminology should not be a privilege, yet many commercial AI pipelines treat expressive fidelity as a premium feature. For AAC users, each mispronunciation requires additional labor—manual corrections, phonetic workarounds, or repeated testing—translating directly into time and cognitive load. These burdens emerge from how TTS systems are trained, evaluated, and commercialized, where frequency, fluency, and market viability often take precedence over expressive variability. The mechanisms below show how technical and economic design decisions translate inequity into infrastructure (see Table 1).

1.3 Inaudible users: data provenance and model assumptions

For many minimally speaking people, the microphone becomes a wall rather than a bridge. Commercial assistants such as Siri and Alexa already underperform for accented speakers (Prinos et al. 2024). But for minimally speaking autistics, whose communication may involve atypical prosody, echolalia, reduced articulation, or speech that is still developing, the failure is more fundamental: the system often fails to recognize their vocalizations as speech at all. What reads as “silence” in the transcript is not an absence of intent; it is an artifact of how intelligibility is defined and detected. This invisibility is engineered across the pipeline.

Recent developments complicate any categorical claim that atypical speech is unrecognized by commercial ASR systems. Google’s Project Euphonia has expanded substantially, now comprising over 1.5 million utterances from approximately 3,000 speakers with disordered speech across multiple languages (Martin et al. 2025). Fine-tuned models within this framework have demonstrably reduced median word error rates for participating speakers. In parallel, recent work using fine-tuned Whisper architectures shows that personalized modeling with limited speaker-specific samples can significantly improve recognition accuracy for dysarthric speech (Raja et al. 2025). These advances indicate meaningful product-level remediation in controlled or enrolled settings.

However, these improvements do not yet represent default pipeline behavior. Large-scale ASR systems remain trained primarily on corpora dominated by normative speech patterns, and personalized fine-tuning remains optional, resource dependent, or unavailable by default. Recognition gains for participating speakers do not imply comprehensive inclusion across the heterogeneity of minimally speaking autistic communicators, whose articulation patterns may vary dynamically across context, state, fatigue, and co-occurring motor conditions. The structural concern, therefore, is not whether recognition can be improved in principle, but whether atypical articulation is treated as a baseline design assumption in dataset construction, evaluation benchmarks, and deployment defaults.

Although few datasets are explicitly labeled by neurotype, the functional effect is the same: speech exhibiting atypical rhythm, articulation, or prosody remains absent or underrepresented. This gap differs from, but compounds, other documented disparities such as accent and dialect bias (Koencke et al. 2020; Liu et al. 2022; Lopez Lloreda 2020). Where disability-related speech does appear, it is typically through clinically labeled corpora that frame atypical prosody or articulation as diagnostic deviation to be *identified and corrected* rather than communicative variation to be *modeled and learned from* (Shor et al. 2019). Model assumptions

then harden those biases. Acoustic front ends and attention mechanisms are optimized for steady temporal and spectral features, penalizing variability in rhythm, articulation, or pause structure (Qian et al. 2023). On the language side, LLM objectives reward statistical predictability: token sequences that deviate from normative linguistic patterns—repetition, mixed modalities, non-linear phrasing—may be down-weighted during training (Rae et al. 2021). For users whose communication diverges from these expectations, deviation itself becomes inaudibility. The problem is not only higher transcription error but outright non-detection. In many cases, the system fails to detect or reliably transcribe the speaker.

A similar hierarchy operates in multilingual AI. LLMs systematically prioritize English and other high-resource languages, even when prompted in lower-resource ones, producing distorted or incomplete outputs (Sharma et al. 2024). This linguistic stratification mirrors prosodic stratification in speech: what is most common becomes legible; what is rare becomes noise. Just as low-resource languages are treated as secondary, minimally produced or atypically patterned speech is sidelined, reinforcing which voices AI learns to recognize, and which it learns to ignore. These failures accumulate from data collection through evaluation. They are not isolated anomalies but reproducible outcomes of how training corpora and objectives are structured (see Table 1).

1.4 Developmental arc assumes static

Most AI evaluation pipelines treat communication as a snapshot rather than a process. A speech or language model is typically validated on a fixed corpus—one utterance, one condition, one “ground truth” label. But for minimally verbal autistics, communicative capacity is dynamic, not static. Expression may expand or contract depending on regulation state, environment, motor fatigue, and predictability (Dinstein et al. 2012; Torres et al. 2013). Static testing regimes obscure these fluctuations, misrepresenting adaptability as inconsistency and momentary silence as incapacity.

Beyond short-term variability, many nonspeaking and minimally speaking autistics also follow developmental trajectories in which expressive capacity may increase over the years, particularly when given access to robust AAC training and responsive support (Brady et al. 2016; Kasari et al. 2014; Schreibman et al. 2015). Some autistic individuals once considered “nonspeaking” later acquire reliable forms of expressive communication; text, symbolic, or spoken, when sensory, motor, and social demands are accommodated (Iacono et al. 2018). Yet AI systems and evaluation pipelines rarely model this longitudinal change. When speech recognition, predictive text, or AAC suggestion algorithms are tuned only to baseline data, they fail to recognize developmental progress and may even penalize improvement as

inconsistency. A model trained on an early communication sample may later misclassify new or more complex utterances as noise or error. Across both timescales, the immediate state and the long arc of development, AI systems are not designed to learn with the user. Current benchmarks optimize for performance on static datasets: one session, one score. Longitudinal variation, session-to-session drift, and learning curves are largely absent from AI evaluation protocols (Masina et al. 2020). As a result, systems are tuned for momentary accuracy rather than sustained usability. The same model that performs well in a calm, predictable setting may fail entirely during fatigue or sensory overload, but those failures are invisible in benchmark reporting.

Accounting for change over time—both situational and developmental—requires rethinking what counts as stability in accessibility research. For populations whose expressive bandwidth naturally fluctuates or expands with practice, true inclusion depends on adaptability. Systems must calibrate to users across states, not just at a single point in time, and evolve with them as communication grows. Ignoring that trajectory encodes a “neurotypical time constant” into design, one that assumes communication is linear, fluent, and temporally stable. For many autistics with communication challenges, it is none of these things. These design assumptions manifest directly in evaluation practices that treat communication as fixed rather than evolving (see Table 1).

1.5 Feedback loops of invisibility

When systems fail, users adapt or withdraw, and the data interpret their departure as satisfaction. If a voice interface fails a user, autistic users often abandon the feature or switch to alternative modalities, if available. Yet telemetry logs that withdrawal as “inactive,” not “excluded.” Aggregate accuracy remains high even as an entire group churns out quietly. The system appears fair precisely because its failures are invisible (Hovy and Spruit 2016). This “invisibility loop” is particularly acute for minimal and nonspeaking autistics, who already carry greater communication labor to work around systems not built for them; initiating, sequencing, and executing motor actions demand sustained cognitive and motor control and, on average, yield much slower output rates than natural speech (often ~10–15 words per minute vs. ~150–200 words per minute for speech for letter by letter selection) (Cler et al. 2019; Trnka et al. 2008; Vargas 2019). Text or symbol selection can demand sustained planning, conscious sequencing, and motor regulation, each layer adding friction to already effortful communication. When technology compresses this diversity of communicative pace and modality into narrow templates of legibility—where “successful communication” equals fast, fluent, and speech-like output—it risks erasing slower or alternative expressions from what counts as meaningful communication.

What looks like user “choice” or “low engagement” may, in practice, be quiet attrition: users leaving not because the technology meets their needs, but because it fails them too completely to continue.

Unless product analytics explicitly track silent abandonment, fallback behavior, and group-specific error distributions, the pipeline records success where users experience failure (Masina et al. 2020). Even “inclusive” benchmarks can mask inequity by defining success through aggregate scores that privilege one canonical way of speaking, text, or reading. The result is a feedback loop of invisibility: the less legible a group becomes, the less data is collected to improve their experience, further reinforcing invisibility in the next design cycle. The fix is not to abandon evaluation, but to redefine what counts as success. True accessibility metrics should measure persistence, not just participation, tracking who remains able to use the system after initial failure, not merely who logs in once. This feedback loop is sustained through how system performance is measured (see Table 1).

1.6 Single-axis fairness

AI fairness frameworks often treat identity as a checkbox, not an ecosystem. Accessibility testing is typically single-axis, focused on one identity category at a time, such as “disability” or “culture,” without examining how those dimensions interact. Vendors may test for “blind” or “deaf” accessibility in isolation, but rarely for intersectional combinations, such as a deaf–blind nonspeaking user relying on tactile input, or a multilingual autistic user navigating both prosodic and linguistic variability. The absence of intersectional evaluation frameworks means that even well-intentioned accessibility audits can reproduce able-bodied and neurotypical defaults. This “fairness by fragmentation” embeds structural erasure into the design process itself (Benjamin 2019; Costanza-Chock 2020).

The result is a paradox: AI systems built in the name of inclusion often fail precisely at the intersections where access is most needed. For instance, datasets labeled as “accessible” might include captions for deaf users or alternate text for blind users, but exclude those who rely on symbol-based AAC or tactile gestures, whose input patterns do not fit the system’s predefined accessibility schema. When fairness frameworks disaggregate categories rather than recognize their interdependence, the burden of adaptation falls back on the user, not the technology. True inclusion requires a shift from accommodation to mutual adaptation: systems that learn from and adjust to users’ diverse communication ecologies. When AI learns from multiple sensory-motor and linguistic pathways, it does not merely “include” disabled users; it broadens collective understanding of what counts as communication. Difference is not deviation but information:

variability itself becomes a design signal rather than a source of error (Masina et al. 2020; Torres et al. 2013).

Designing for intersectional accessibility is not simply an ethical add-on; it is a methodological imperative. Systems that ignore intersectionality produce narrow generalizations that fail under real-world diversity. Conversely, systems designed with intersectional inputs produce more robust, generalizable models. Designing for mutual adaptation transforms accessibility from a compliance exercise into a driver of innovation. Yet when inclusion is framed primarily as innovation, it risks being measured by its economic or productivity gains rather than its human impact. Furthermore, inclusion must not be reduced to productivity or market value. AI systems that commodify neurodivergent “strengths” such as attention to detail, deep focus, or pattern recognition risk recasting disability as a form of conditional belonging, valuable only when profitable. A disability-justice approach to technology design begins not with efficiency but with dignity and agency (Costanza-Chock 2020; Piepzn-Samarasinha 2018). It reorients the design question from *How can disabled people fit our systems?* to *How can our systems expand to fit disabled lives?* The gap between inclusion rhetoric and real-world testing can be traced to how fairness itself is operationalized (see Table 1).

1.7 AI plain-language generation

Plain language is an important accessibility tool for the disability community, including for many minimally and nonspeaking autistics. Plain language is the distillation of complex language into more layman language. The intent of plain language is to ensure that readers can find, understand, and use information the first time without altering its meaning, tone, or intent (Plain Language Action & Information Network n.d. 2026; U.S. Office of Management and Budget 2011). It enables someone to understand a medical form, a benefits letter, or a research summary that would otherwise remain locked behind technical or bureaucratic jargon. For many disabled readers plain language is the bridge between information and participation. For AAC users, plain language can also be the bridge between reception and expression. Dense or ambiguous text places additional cognitive and motor demands, compounding the inequities already created by inaccessible design. Plain language is therefore not simplification for its own sake; it is a mode of communication built on respect for comprehension and reciprocity. At its best, plain language is not condescension but conceptual fidelity delivered with care.

In recent years, AI has become the main source of plain-language generation. LLMs now promise to instantly “translate” any text into an “easy-read” or “accessible” version. Government agencies, hospitals, and even disability organizations increasingly rely on these tools to

scale communication. The appeal is obvious: speed, consistency, and lower cost. But what these systems optimize for and what they erase reveal a deeper problem. To an algorithm, “plain” often means shorter simpler sentences; to a disabled reader, it means accurate, respectful, and usable. When AI confuses brevity with accessibility, fidelity becomes the casualty. AI-generated plain language may support comprehension in routine or low-stakes contexts, such as navigation prompts or appointment reminders, but it is ill-suited to rights, consent, or benefits communication unless reviewed by human experts familiar with disability discourse. Simplification systems smooth the rough edges of complexity but also erase intent and perspective. These same dynamics underpin the generative models that increasingly power voice assistants and plain-language tools alike. Systems that convert between text and speech share the same training data and value assumptions about who is “clear” or “coherent.”

This drift begins upstream. LLMs are trained predominantly on text authored by sighted, hearing, and nondisabled writers, entrenching majority norms for what counts as “coherent,” “neutral,” or “clear” (Bender et al. 2021). When plain-language training corpora are assembled without disabled readers’ perspectives, accessibility is misdefined as short sentences rather than faithful meaning (Alva-Manchego et al. 2020). Downstream, optimization for readability further tilts models toward fluency at the expense of meaning. Simplification and summarization systems fine-tuned on metrics such as sentence length, syntactic simplicity, and word frequency can distort semantics—particularly around disability, benefits, or legal language—precisely where nuance matters most (Alva-Manchego et al. 2020; Freyer 2024). For example, a model might *hypothetically* reframe “sheltered workshops,” segregated workplaces where disabled people are paid subminimum wages, as positive, simply because the word “shelter” connotes safety (Freyer 2024). Such outputs are not harmless errors; they constitute epistemic harm disguised as clarity. Meaning shifts, trust erodes, and access narrows. For communities that depend on plain-language content to engage in civic and institutional life, misrepresentation in wording can mean exclusion from rights, benefits, or informed consent. The failures are not solely technical but epistemological, rooted in whose words and worldviews train the system. Most text corpora encode majority cultural and cognitive norms, defining “readable” and “neutral” through nondisabled expectations of language and tone (Bender et al. 2021). Bias begins at data provenance, long before a single output is generated. Addressing these failures therefore requires reframing plain-language AI not as a convenience feature but as a site of justice and consent. Asking, “How can we make AI write this?” misses the point. The real question is: *Who should be leading this work—and who decides when meaning has been lost?*

Two mechanisms drive these distortions: (a) semantic loss, where essential disability or policy terms drop out during simplification, and (b) stance drift, where value-laden phrases are reinterpreted through majority linguistic norms. Each demands different safeguards; terminology preservation for the first, perspective auditing for the second. These distortions are traceable to the way corpora and objectives are structured (see Table 1).

1.8 Agency: technology that speaks about us, not with us

When meaning shifts, so does agency. The issues seen in plain-language generation, where AI decides how to “simplify” information for disabled readers, mirror a broader pattern across health, education, and communication technologies: systems that speak about disabled people, not with them. Whether a voice interface, an AAC program, or an AI-powered classroom tracker, the default assumption is that expertise and consent reside elsewhere, with “the other adult in the room.” Indeed, in practice, many tools are designed to interact *through* caregivers, clinicians, or teachers rather than *directly* with disabled individuals. AAC systems are often configured and updated by speech therapists, educational staff or other caregivers, with limited input from the user who must live with the interface daily. Similarly, AI-based “engagement” or “progress tracking” systems used in classrooms and clinics monitor autistic students’ facial expressions, gaze, or motor activity, sending reports to instructors rather than empowering the students themselves (Alper 2017; CDT 2022). The result is agency by proxy: the technology becomes another layer through which others interpret and manage the disabled person, rather than a tool of self-expression or autonomy.

As disability scholars have noted, many assistive technology paradigms historically emerged from clinical models emphasizing supervision and correction rather than autonomy (Hamraie 2017; Hendren 2020). This caregiver-centric orientation often extends beyond design itself—into deployment and institutional control—shaping how communication technologies are used in classrooms, clinics, and research contexts (Hamraie 2017). Assistive technologies have long been built around the *paternalistic* assumption that the user must be interpreted, monitored, or managed. For minimally or nonspeaking autistics, these assumptions often take concrete, embodied forms. Devices presume steady gaze, clear articulation, and precise hand control. Interfaces expect rapid touch or eye-tracking input, even though many users experience apraxia, tremors, or inconsistent motor control. Some rely on rhythmic movement or gesture to regulate attention or sensory load—behaviors that remain poorly represented in the datasets used to train sensing and interaction models

(Torres et al. 2013). Even with AAC access, communication for many nonspeaking users remains slower and more effortful than speech, requiring sustained planning and motor regulation (Cler et al. 2019; Trnka et al. 2008; Vargas 2019).

Yet design and analytics frameworks seldom account for this temporal diversity. By optimizing for fluency and speed, many systems implicitly equate “successful communication” with speech-like output, marginalizing slower or alternative forms of expression. While disability studies richly theorizes exclusion and dignity, it often stops short of specifying how these dynamics manifest within ML architectures. Conversely, mainstream AI fairness research frequently addresses demographic bias without sustained engagement with disability as a structurally mediated communicative condition.

Agency loss in AI thus operates on multiple levels: *interface*, *inference*, and *institution*. At the interface level, disabled users are excluded from configuring their own tools. At the inference level, AI systems interpret bodily and communicative signals through nondisabled norms. At the institutional level, deployment contexts (schools, clinics, research laboratories) continue to vest control in professionals rather than participants. The cumulative effect is a system that grants the illusion of assistive empowerment while re-inscribing dependency. A truly equitable AI ecosystem would invert this hierarchy: putting the autistic user in the driver’s seat. That means interfaces co-designed with disabled users, data pipelines that include their feedback loops, and governance structures that give them veto authority over how their data and likeness are used. As long as AI systems continue to “speak for” rather than “with,” inclusion will remain conditional, contingent on compliance with neurotypical norms. However, the question of agency demands both care and realism. Many minimally and nonspeaking autistics remain without consistent AAC access or instruction, limiting their ability to participate even in decisions about their own care or technology use (Light and McNaughton 2012, 2013). In such contexts, agency cannot simply be presumed—it must be scaffolded. Families and caregivers often act as intermediaries, voicing needs or preferences on behalf of the AAC user. This dynamic can be both protective and constraining: it reflects genuine care and necessity in the absence of reliable communication systems, yet it risks reinforcing dependency and erasing the individual’s evolving voice. “Nothing About Us Without Us” thus requires a complementary notion of *co-agency*; designing systems that allow shared control between users and trusted partners. Ethical design should record proxy actions transparently, permit retrospective user review, and support gradual transfer of control as expressive capacity evolves (Costanza-Chock 2020; Light and McNaughton 2012). The challenge of agency is thus not merely ethical

but architectural. How tools are configured determines who holds control (see Table 1).

Recent technological advances complicate a purely deficit-oriented account. Commercial ASR systems have expanded atypical speech corpora; personalized fine-tuning has improved word error rates for some dysarthric speakers, and voice-cloning tools have reduced barriers to identity-matched synthesis, once prioritized. These developments demonstrate that exclusion is not technically inevitable. Yet, these improvements often operate as optional layers—personalization features, premium tiers, or research deployments—rather than as default baselines within mainstream pipelines. Distributional assumptions about normative speech, stable articulation, and majority-language usage continue to structure model training and evaluation. In this sense, product-level remediation can coexist with infrastructural exclusion. The persistence of these underlying design assumptions motivates the governance analysis that follows.

2 Governance and accountability: toward designed dignity

If engineered exclusion describes how access fails, designed dignity asks how it can be built in. Governance, in this context, extends beyond compliance or regulation—it encompasses the full ecology of decision-making that determines who is included, heard, and empowered across the AI pipeline. Governance operates across three interconnected levels—external regulation, internal accountability, and participatory co-governance. This broader framing aligns with work in responsible innovation emphasizing anticipatory and value-sensitive embedding of ethics into technological design (van Grunsven and Roeser 2022). External regulation includes policy instruments, auditing standards, and accessibility compliance mechanisms that create enforceable guardrails. Internal accountability refers to institutional design processes such as inclusive review boards, ethical AI evaluations, and accessibility testing embedded within product lifecycles. Participatory governance brings disabled and nonspeaking users into shared decision-making roles, ensuring that access is not merely implemented but co-defined (Benjamin 2019; Costanza-Chock 2020; Hamraie 2017). Governance is thus the mechanism that translates principle into practice, embedding accountability across the AI lifecycle, from dataset creation to model deployment (see Table 2). For minimally and nonspeaking autistics, whose communication is often mediated through technology, governance must do more than mitigate bias; it must safeguard affordability, agency, and continuity of participation even when communication itself is fragile or state dependent.

Regulatory frameworks such as the *EU Artificial Intelligence Act* (European Parliament and Council of the

European Union 2024) and the US Rehabilitation Act, Section 508 (U.S. General Services Administration 2023) already recognize accessibility as a compliance requirement, yet few specify measurable communicative-equity standards. Designed-dignity metrics could extend these models by embedding expressive-fidelity and longitudinal-variability measures into certification checklists. Projects like Google's *Project Euphonia* (Martin et al. 2025), which partnered with dysarthric speakers to improve ASR inclusivity, demonstrate how participatory data collection can align technical and ethical progress. Recent overviews of dysarthric speech technologies similarly note that while model performance has improved under controlled personalization conditions, integration into mainstream commercial ASR pipelines remains uneven (Bhat & Strik 2025). Emerging research comparing idiosyncratic speaker modeling to normative speech modeling further illustrates the tension between personalization and baseline inclusion (Raja et al. 2025).

2.1 From inclusion promises to enforcement mechanisms

Most current AI ethics frameworks emphasize fairness, transparency, and privacy but treat accessibility as a secondary or voluntary domain (Whittaker et al. 2021). This omission has tangible consequences. When accessibility audits are optional, vendors can claim inclusion without ever testing whether nonspeaking or AAC users can operate their systems. A governance model grounded in designed dignity would make accessibility both measurable and enforceable. This means integrating access metrics, like expressive intelligibility, input–output parity, and error recovery rates—into the same compliance pipelines that monitor bias and safety (Masina et al. 2020). The diagnostic metrics outlined throughout “*Engineered Exclusion in Practice*” provide a foundation for this integration: collectively, they define measurable points where accessibility can be audited, monitored, and improved. Accessibility cannot remain an “after-market” concern; it must be a regulated design requirement with traceable accountability. Ensuring such accountability, however, also requires representation in how those standards are defined and enforced.

2.2 Participatory and tiered oversight

Because not all AAC users can reliably self-advocate, governance structures must support tiered representation: direct participation where possible, and proxy participation, through trained family members or trusted staff, where necessary. In participatory AI, users, caregivers, and disability scholars collaborate at every stage: defining datasets, labeling content, interpreting outputs, and deciding what constitutes harm (Costanza-Chock 2020; Light and

McNaughton 2012, 2013). For higher-support autistics, co-agency mechanisms such as consent dashboards, review loops, or digital logs can document decisions while preserving the possibility of future autonomy. Ethical governance should also protect against permanent proxying: caregivers may represent users when communication is limited, but systems must remain adaptable so that as communicative capacity develops, through therapy, assistive technology use, or cognitive change, control over decisions and data access transitions back to the individual. Designed dignity thus frames governance not as static compliance but as temporal stewardship, supporting people through evolving trajectories of voice and agency. Ultimately, designed dignity reframes governance as a living process, one that evolves alongside the user, ensuring that technology serves not only present needs but future autonomy.

2.3 Transparency and traceability

Transparency is often invoked as a principle but rarely extended to disabled users. For accessible AI, transparency means that users and their families should be able to see, and, where possible, edit, how their data, voices, and interaction logs are used. Traceability requires datasets to carry provenance metadata specifying whether disabled or AAC users were included and whether their contributions were labeled as “error” or “signal.” Without such metadata, audits cannot distinguish inclusion from exploitation. Public registries could list certified models that meet minimum accessibility and inclusion benchmarks, analogous to energy-efficiency ratings in consumer products. Each model's listing could display standardized accessibility metrics, such as expressive intelligibility, input–output parity, and support for nonstandard speech inputs, allowing consumers, institutions, and regulators to compare inclusion performance at a glance. These would not merely signal compliance but also drive market competition toward inclusion. Such registries, however, must be grounded in rigorous auditing frameworks that measure not only whether systems can *hear* users, but also whether they can *speak with* them—capturing the full reciprocity of communication.

2.4 Auditing for expressive access

Accessibility audits typically test whether information can be received. For nonspeaking and minimally speaking autistic users, equally vital is whether information can be expressed. A designed-dignity audit must therefore assess expressive pathways: (a) Can users produce speech output through AAC or symbol-to-speech systems at parity with speech-input accuracy? (b) Are users able to correct, override, or personalize automated simplifications or voice selections? Audits that capture only comprehension

reproduce the same asymmetry the paper critiques: they hear input but ignore output.

2.5 Embedding dignity in design culture

Governance is not only procedural but cultural. Metrics and audits matter only when teams internalize that accessibility is the design brief, not an accommodation. Autistic anxiety often arises not from intrinsic fragility but from environments that chronically misread or constrain communication—a dynamic that technological systems risk amplifying when they normalize exclusionary defaults (Srinivasan 2025a). Embedding dignity in design thus requires addressing these relational and structural misattunements, not merely their technical manifestations. Genuine inclusion requires an antinormative shift—one that treats access not as compliance but as a challenge to whose ways of knowing and communicating are centered (Srinivasan et al. 2025). Training programs for data scientists, clinicians, and product managers should include disability-led modules on communication diversity, agency scaffolding, and participatory ethics (Benjamin 2019; Costanza-Chock 2020). Funding agencies and journals can reinforce this shift by requiring accessibility impact statements alongside bias or privacy statements. These statements move inclusion upstream—positioning disabled collaborators not as compliance checkpoints but as contributors who shape problem definitions, data protocols, and evaluation criteria, and who co-interpret outcomes. In this way, designed dignity reframes governance from gatekeeping to co-creation: an ecosystem where disabled users become not passive beneficiaries but co-authors of technological possibility.

2.6 Limitations

This paper centers minimally and nonspeaking autistic users within English-dominant AI systems. It does not address visual-tactile modalities, non-autistic disabilities, or non-Anglophone infrastructures where training data and accessibility standards differ. Claims about “neurotypical” bias are inferred from dataset composition and benchmarking norms, not direct neurotype labeling. The designed-dignity metrics proposed in Table 1 are therefore conceptual and operational in intent, rather than empirically validated standards. Future research should systematically test these metrics across populations, contexts, and languages, alongside mapping disability representation, comparing open-source and commercial systems, and extending analysis to Global South contexts where both data regimes and definitions of accessibility may differ.

3 Conclusion

Artificial intelligence systems mirror the assumptions of their makers: what counts as data, whose communication is legible, and what errors are worth fixing. For minimally and nonspeaking autistics, these assumptions translate into built-in exclusion—when fluency, speed, or predictability are treated as prerequisites for participation. Across speech recognition, text-to-speech, plain-language generation, and analytic feedback loops, the pattern is consistent: systems that privilege the statistically common, systematically erase the communicatively diverse.

This paper has argued that such erasure is not accidental but *engineered*. The inverse, *designed dignity*, is equally engineered: it requires deliberate technical, ethical, and governance choices that build reciprocity, accountability, and co-agency into AI’s infrastructure. Building for expressive access, longitudinal variability, and multimodal communication strengthens, not weakens, scientific rigor. When models learn from a wider range of human communication, they become more adaptive, generalizable, and ethically defensible.

Inclusion must therefore move upstream: into how data are sourced, how meaning is validated, and who decides when a system “works.” Engineers, policymakers, and AAC users alike must share authority in defining success and harm. Accessibility cannot remain an afterthought or a premium feature; it is foundational to fairness and to AI’s claim to universality. If engineered exclusion reflects the limits of today’s systems, designed dignity offers a blueprint for what technology can become: not merely responsive to difference, but responsible to it.

Translating designed dignity into practice requires aligned action across research, regulation, and design. Regulatory frameworks should treat accessibility failures as algorithmic bias, subject to audit and remedy; and AI benchmarks should include expressive fidelity and longitudinal variability as core metrics. In research, centering the needs of minimally and nonspeaking autistics can reorient what counts as valid signal and success. Such shifts ensure that accessibility is not a retrofit, but an organizing principle for responsible AI.

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